

Assignment - Implement Authentication & Authorization

Objective

Enhance the existing **StudentRegistration WebApp** by implementing **Authentication**, **Authorization**, and **Role-Based Navigation** using **ASP.NET Core Identity**.

Requirements

Part 1 - Authentication

1. Enable **Individual Accounts Authentication** for the application (if not already configured).
2. Create an **Administrator** account manually.
3. Allow new users to **Register** and **Login** using ASP.NET Core Identity.
4. After successful registration, every new user should automatically be assigned the **Student** role.
5. After successful login, display the logged-in user's email in the navigation bar.

Part 2 - Authorization

Implement role-based authorization with the following requirements:

Administrator

The Administrator should be able to:

- View the list of all Courses.
- Add a new Course.
- Edit Course details.
- Delete a Course.

Student

View the list of all registered Students.

A Student should be able to:

View the list of available Courses.

Register for **only one** Course.

View his/her own profile.

Edit personal details.

Logout from the application.

A Student **must not** be able to:

- Create, Edit or Delete Courses.

View details of other students.

View the complete Student List.

Part 3 - Navigation Menu

Display navigation links based on the logged-in user's role.

Anonymous User

- Home

Courses

Register

Student

Login

Home

Courses

Register for Course (*display only if the student has not already registered for a course*)

My Profile

Logout

Administrator

- Home

Courses

Students

Logout

Part 4 - Security

Apply appropriate authorization attributes to secure the application.

Anonymous users should be able to view the Home page and Course List.

- Only authenticated Students should be able to register for a course.

Only Administrators should be able to manage Courses.

Users should not be able to access restricted pages by typing the URL directly.

Part 5- User Interface

Display a welcome message after successful login.

- Display the logged-in user's email on the navigation bar.

Show appropriate messages for successful registration, login, logout, and course registration.

Display an **Access Denied** page whenever a user attempts to access an unauthorized page.

Expected Functionality

After completing the assignment, the application should satisfy the following workflow:

Anonymous User

Browse Courses

Register Login

Student Log in

Register for Course
View My Profile
Edit My Profile Logout

Administrator Login

Manage Courses
View Student List
Edit/Delete Courses Logout

Submission Requirements

Students should submit:

Source Code of the project (**separate .cs file** or in a **single pdf or .docx**).

Screen shot of SQL Server database containing Identity and application tables.

A screenshot of UI:

- Administrator Navigation

- Student Navigation

- Anonymous Navigation

- Successful Login

- Successful Course Registration

- Access Denied page

- Student Profile page

This assignment naturally extends the previous database and MVC assignment into a complete role-based web application while reinforcing Identity, Authentication,

Authorization, and role-based UI customization.